

Olek Osikowicz

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Education

- PhD** **University of Sheffield**, School of Computer Science Sheffield, UK
• *Efficient and reliable simulation-based Autonomous Driving Systems testing* Sept 2023 – Sept 2027
• PhD Supervisors: *Donghwan Shin & Phil McMinn*
- BSc** **University of Sheffield**, Computer Science Sheffield, UK
• Graduated with first-class honours, Dissertation: *Grounded In Reality Autonomous Driving Systems Testing* Sept 2020 – June 2023
• Supervisor: *Phil McMinn*

Autonomous Driving Testing

Accelerating Autonomous Driving Evaluation

- Accomplished 16.8% cost reduction in finding critical failures, measured by total effectiveness over CPU-hours
- Achieved by designing Multi-Fidelity Bayesian Optimization algorithm for testing complex black-box systems
- Project supported by £58K NVIDIA Drive academic grant to evaluate in Software-in-the-Loop settings

Software- and Hardware-in-the-Loop Testing

- Led a two-person team to develop a bridge between Autoware stack (C++ / ROS) and CARLA simulator
- Deployed Autonomous Driving stack on NVIDIA DRIVE AGX Thor Dev Kit for hardware-in-the-loop (HIL) testing
- Enabled deterministic replays of driving scenarios to maximize test reproducibility and reliability in CARLA

Scalable ML Infrastructure

- Reduced experiment execution time from months to 1 week by parallelizing model evaluations
- Achieved by deploying a multi-node computing cluster using Ray actors & AWS EC2 to distribute agent evaluations
- Engineered real-time pipeline health monitoring with Prometheus, Loki, and Grafana services

Skills

Autonomous Driving Tooling: Autoware, CARLA, ROS/ROS2, AlpaSim, OmniDreams, Neural Reconstruction (NuRec)

Software Engineering: Python, C++, Data Structures & Algorithms, Git, CI/CD, Unit & Integration Testing, System Design

Performance & Observability: Profiling (cProfile), Async Programming, Ray, Docker, AWS, Prometheus, Grafana

ML Tooling: PyTorch, Hugging Face, Generative AI (VAEs, World Models, Transformers), Bayesian Optimization

Distributed computing: Ray, Multiprocessing, Docker, AWS (EC2/S3), SLURM, Loki, Prometheus, Grafana

Publications

Simulation Nondeterminism and Its Impact on Flaky Tests for Autonomous Driving Systems Oct 2026

Olek Osikowicz, Phil McMinn, Donghwan Shin

Accepted at *Automated Software Engineering (ASE) Journal*

DT-Drive: A Tool for Deterministic Replay-Based Testing and Debugging of Autonomous Driving Systems Sept 2026

Sanjeetha Pennada, Matthew Leach, Olek Osikowicz, Donghwan Shin

Accepted at *2026 International Conference on Automated Software Engineering (ASE 2026)*

Multi-Fidelity Bayesian Optimization for Simulation-Based Autonomous Driving Systems Testing	June 2026
Olek Osikowicz, Phil McMinn, Wei Xing, Donghwan Shin 10.1109/IV66570.2026.11623850 🔗 2026 IEEE Intelligent Vehicles Symposium (IV 2026)	
CAWSR: Carla-AutoWare Scenario Runner	June 2026
David Gasinski, Olek Osikowicz, Gwilym Rutherford, Donghwan Shin github.com/Intelligent-Testing-Lab/cawsr 🔗 Under review at <i>Journal of Open Source Software (JOSS)</i>	
Empirically Evaluating Flaky Tests for Autonomous Driving Systems in Simulated Environments	Apr 2025
Olek Osikowicz, Phil McMinn, Donghwan Shin 10.1109/FTW66604.2025.00009 🔗 2025 IEEE/ACM International Flaky Tests Workshop (FTW 2025)	

Employment

University of Sheffield , Research Assistant in Simulation-Based Testing	Sheffield, UK
- Collaborating with KAIST (Korea Advanced Institute of Science and Technology) - Led 2 researchers to develop a state-of-the-art C++ dockerized driving deployment	June 2025 – June 2026
Dover Fueling Solutions , Data Engineering Intern	Kraków, Poland
- Built batched ETL data pipelines on MS Azure to process & clean large-scale datasets - Joined datasets via SQL warehouses on Databricks supporting analytics & modelling	June 2022 – Sept 2022

Teaching

Software Re-Engineering 🔗	Mar 2024 – present
Led classes for 3rd and 4th year students in re-engineering real-world Python projects	
Introduction to Algorithms and Data Structures 🔗	Feb 2023 – June 2023
Ran tutorial sessions for first-year students, explaining core data structures and algorithmic complexity (Big O notation)	

Awards

NVIDIA Academic Grant (Robotics & Edge AI)	Mar 2026
- Supporting MF-Drive project addressing the critical trade-off between scalability and fidelity in AV testing - Hardware valued £58,000: NVIDIA DRIVE AGX Thor, RTX PRO 6000 Blackwell, and 32K A100 GPU-hours	
Best teaching support award by School of Computer Science	July 2025
- Recognition of excellence in teaching support as a Graduate Teaching Assistant in Software Re-engineering - Awarded by Head of School in the 2024/25 academic year	
University of Sheffield Billy Ibberson Bursary	Feb 2025
- Secured a competitive £1,771 grant for travel expenses and research training - Awarded to support research that develops links between academic innovation and industrial practice	
Software Hut Prize	June 2023
- Awarded for the best software delivery to a real-world client - Recognized for excellence in Agile development and effective team collaboration	

Service

Facilitator, Alan Turing Institute PhD Student Presentations & Networking Day	Feb 2026
Assisted in organising the event, welcomed guests, and moderated Q&A sessions following each presentation	
Reviewer, Artifact Evaluation Committee Member for ESOP, FASE and FoSSaCS	Jan 2026
Reviewed and evaluated research artifacts for reproducibility	
Student Volunteer, International Conference on Software Engineering (ICSE 2025)	Apr 2025
Assisted in organizing and managing conference sessions and events	